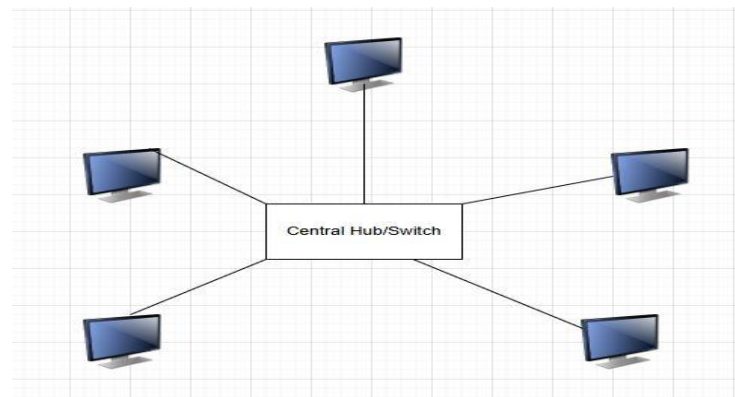


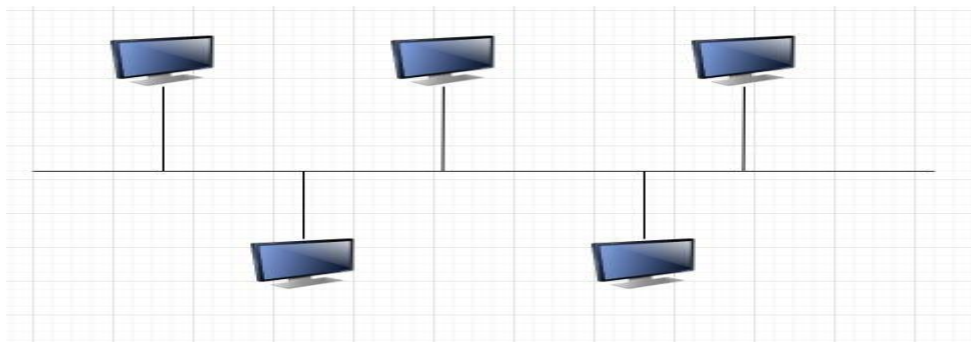
1. Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 days.

1. Star Topology



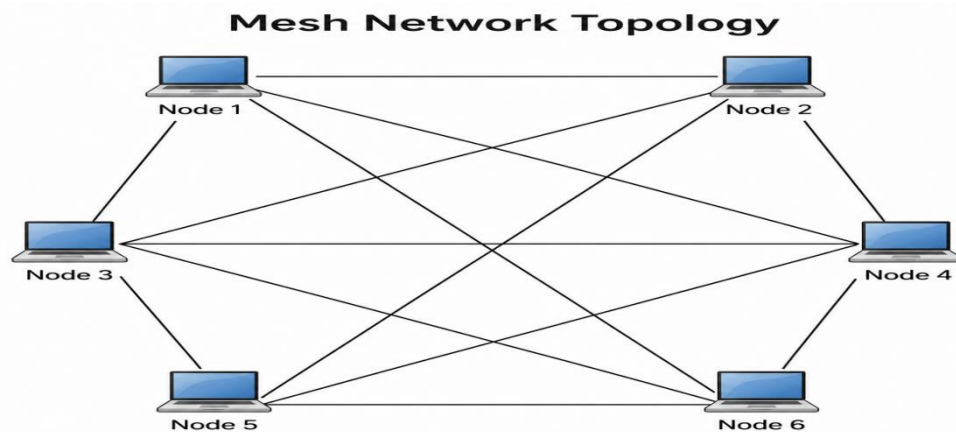
- **Real-world App/service: WhatsApp**
- WhatsApp uses a central server architecture where all clients connect to centralized server to send or receive messages.

2. Bus Topology



- **Real-world App/service: IPL Live Streaming (Jio-Hotsar)**
- Content is distributed over a shared network backbone to many users simultaneously, making bus topology suitable for one-to-many data delivery in a network.

3. Mesh Topology



- **Real-world App/service : Flipkart**
- Flipkart's services such as search, cart, payment, inventory and logistics need high availability and fault tolerance. Mesh topology provides multiple paths, so the system stays reliable even if one link/server fails.

2. Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use.

- Here is comparison of different topologies:

1. Star Topology

- **Pros:**

1. Easy to Install and manage.
2. Failure of one device does not affect others.
3. Easy to detect the faults.
4. Good performance because of dedicated connections.

- **Cons:**

1. Central hub/switch is a single point of failure.
2. Requires more cable than bus topology.
3. Higher Installation cost.
4. Network depends heavily on the central device.

- **Best Use Case:**

- Homes, schools, offices and most modern LANs.

- **Example:**
- Instagram – Data from your phone connects through central servers, similar to devices communicating through a central switch.

2. Bus Topology

- **Pros:**
 1. Low installation cost.
 2. Requires less cable.
 3. Easy to set up for small networks.
 4. No central device required.
- **Cons:**
 1. Backbone cable failure brings down the entire network.
 2. Performance decreases as more devices are added.
 3. Difficult to troubleshoot.
 4. Limited cable length and number of devices.
- **Best Use Case:**
 - Small temporary networks or simple lab networks.
- **Example:**
 - Zomato – a simple local network in a small restaurant where one shared cable connects billing and order systems.

3. Ring Topology

- **Pros:**
 1. Data collisions are minimized.
 2. Equal access for all devices.
 3. Predictable performance under heavy load.
 4. Can handle higher traffic than bus topology.
- **Cons:**
 1. Failure of one device or cable can disrupt the entire ring.
 2. Adding or removing devices can interrupt the network.
 3. Troubleshooting is more difficult.
 4. Higher setup complexity than bus topology.
- **Best Use Case:**
 - Industrial systems and legacy token-ring environment requiring orderly communication.

- **Example:**
- Paytm – Transaction processing can be compared to data moving in an orderly sequence, although the actual service runs on modern data center networks rather than a ring.

4. Mesh Topology

- **Pros:**

1. Very high reliability.
2. Multiple paths ensure redundancy.
3. Excellent fault tolerance.
4. High security and availability.

- **Cons:**

1. Very expensive to install.
2. Requires a large amount of cabling or wireless links.
3. Complex configuration and maintenance.
4. Difficult to scale in fully connected networks.

- **Best Use Case:**

- Data centers, military networks, critical communication systems.

- **Example:**

- Instagram, Zomato and Paytm – their cloud infrastructure uses redundant connections between servers so services remain available even if some networks links fail.

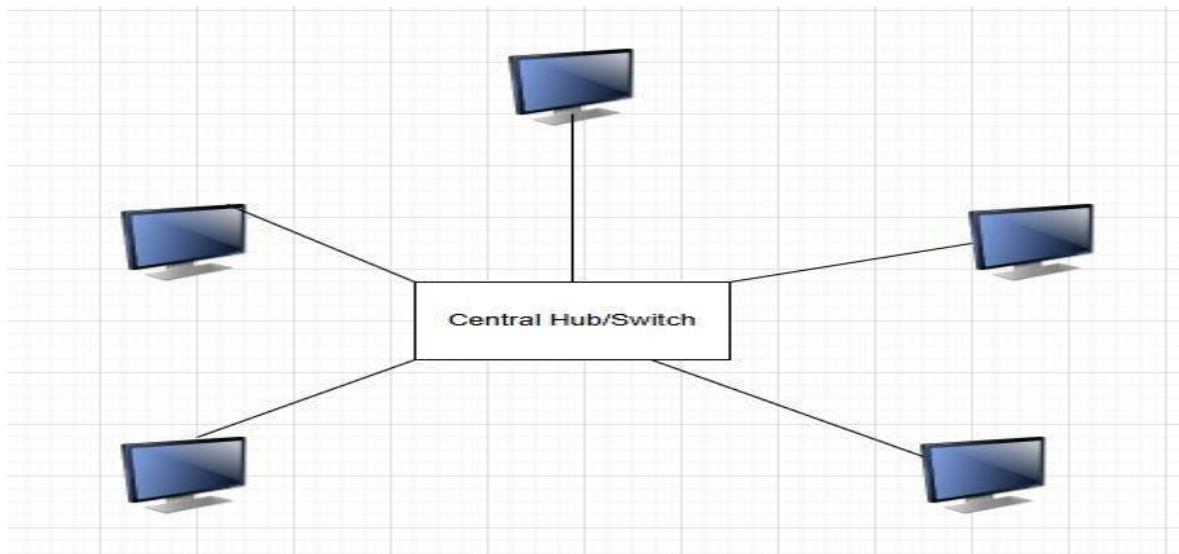
3. Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

- I would choose a client-server topology with geographically distributed edge/region servers. All players connect to the nearest regional servers, which keeps the game state synchronised and prevents cheating better than peer-to-peer networks. This reduces latency and improves game speed for players in different areas, while centralized control increases reliability and simplifies updates and maintenance. The cost is higher than a pure P2P system because it requires multiple cloud servers

and load balancers, but it provides much better scalability and a smoother multiplayer experience.

4. Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines.

- I would use Star topology for this scenario.



- A star topology is highly cost-effective and easy to manage for 8 devices because every computer connects directly to a central switch. If one cable fails, only that single workstation goes offline without interrupting the rest of warehouse operations. Additionally, scaling up in the future is as simple as a new computer into an empty port on the switch.